AFFINE ALGEBRAIC VARIETIES

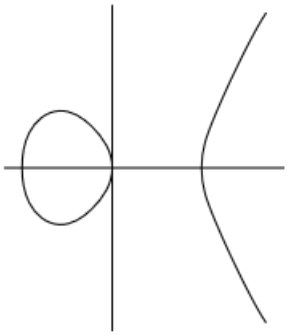
In this lecture, we will cover some basic theory of algebraic varieties. We will not prove all the things covered here as this and the next few lectures are to be taken more like a glossary of terms and results that we will use. The students are expected to work out the details on their own, as this will give them much needed experience with these objects.

1. Affine space and algebraic sets

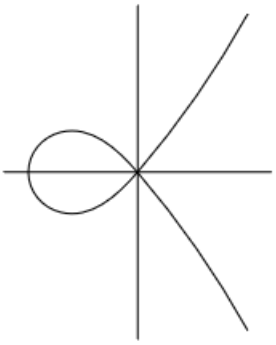
Definition 1.1. Let k be any field. By $\mathbb{A}^n(k)$, or simply $\mathbb{A}^n$ if k is understood, we will mean the cartesian n-fold product of k with itself. (Note that this definition changes when we transfer to scheme theory; but for now this will be enough.) We call $\mathbb{A}^n(k)$ *affine n-space over k*. Its elements will be called 'points'.

If $F \in k[X_1, \ldots, X_n]$, a point $P = (a_1, \ldots, a_n)$ is called a *zero* of F if $F(P) = F(a_1, \ldots, a_n) = 0$. Let k be algebraically closed. If F is not a constant, the set of zeroes of F is called the *hypersurface defined by F* and is denoted as $V(F)$. A hypersurface in $\mathbb{A}^2(k)$ is called an *affine algebraic curve*. If F is a polynomial of degree 1, $V(F)$ is called a *hyperplane* in $\mathbb{A}^n(k)$.

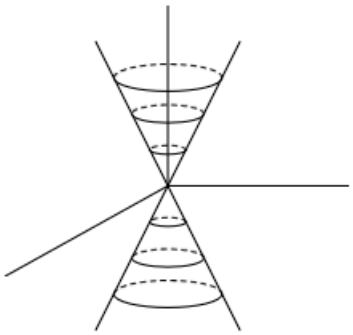
Below are some affine curves and hypersurfaces for $k = \mathbb{R}$.



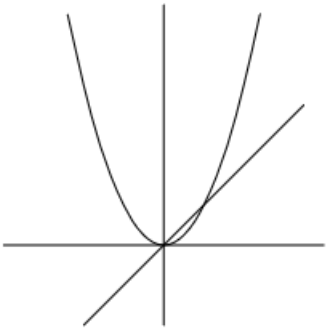
a. $V(Y^2 - X(X^2 - 1)) \subset \mathbb{A}^2$



b. $V(Y^2 - X^2(X + 1)) \subset \mathbb{A}^2$



c. $V(Z^2 - (X^2 + Y^2)) \subset \mathbb{A}^3$



d. $V(Y^2 - XY - X^2Y + X^3)) \subset \mathbb{A}^2$

Definition 1.2. More generally, if S is any set of polynomials in $k[X_1, \ldots, X_n]$, we define

$$V(S) := \{P \in \mathbb{A}^n : F(P) = 0 \text{ for every } F \in S\}.$$

So in particular $V(S) = \bigcap_{F \in S} V(F)$. Note that if $S = \{F_1, \ldots, F_r\}$, we usually write $V(F_1, \ldots, F_r)$ instead of $V(\{F_1, \ldots, F_r\})$.

A set $X \subset \mathbb{A}^n(k)$ is defined to be an *affine algebraic set*, or simply algebraic set if the context is understood, if $X = V(S)$ for some $S \subset k[X_1, \ldots, X_n]$.

In particular, note the following properties that are routine to prove:

Lemma 1.3 (Properties of algebraic sets)**.** *We have the following:*

- *If I_S denotes the ideal generated by S in $k[X_1, \ldots, X_n]$, then $V(S) = V(I_S)$. In particular, every affine algebraic set is of the form $V(I)$ for some ideal I.*
- *If $I \supset J$, then $V(I) \subset V(J)$.*
- *If $\{I_\alpha\}$ is any collection of ideals, then $V(\bigcup_\alpha I_\alpha) = \bigcap_\alpha V(I_\alpha)$; so the intersection of any arbitrary collection of affine algebraic sets is again an algebraic set.*
- *$V(I) \bigcup V(J) = V(\{FG : F \in I, G \in J\}$; so any finite union of affine algebraic sets is again an algebraic set.*
- *$V(0) = \mathbb{A}^n$; $V(X_1 - a_1, \ldots, X_n - a_n) = \{(a_1, \ldots, a_n)\}$ so that any finite subset of $\mathbb{A}^n$ is an affine algebraic set.*

Definition 1.4. Let X be any affine algebraic set in $\mathbb{A}^n$ and let I denote the set of polynomials that vanish on X; we know that this forms an ideal $I \subset k[X_1, \ldots, X_n]$. We call it the *ideal of* X and write it as $I(X)$.

Note that by Hilbert Basis Theorem, all ideals in this polynomial ring are finitely generated, which can be written in a geometric form:

Theorem 1.5. *Every algebraic set is the intersection of a finite number of hypersurfaces.*

We have similar properties as before:

Proposition 1.6 (Properties of ideals of algebraic sets)**.** *We have the following:*

- *If $X \subset Y$, then $I(X) \supset I(Y)$.*
- *$I(\emptyset) = k[X_1, \ldots, X_n], I(\mathbb{A}^n(k)) = \{0\}$ if k is an infinite field. $I(\{(a_1, \ldots, a_n)\}) = (X_1 - a_1, \ldots, X_n - a_n)$ for $a_1, \ldots, a_n \in k$.*
- *$I(V(S)) \supset S$ for any set of polynomials S; $V(I(X)) \supset X$ for any set of points X.*
- *$V(I(V(S))) = V(S)$ for any set of polynomials S; $I(V(I(X))) = I(X)$ for any set of points X.*

Ideals of algebraic sets have a property that is not true about all ideals of the polynomial ring.

Lemma 1.7. *For any $X \subset \mathbb{A}^n(k)$, the ideal $I(X)$ is a radical ideal.*

As will be the theme throughout algebraic geometry and this course, algebraic properties of $I(X)$ correspond to geometric properties of X.

Definition 1.8. An algebraic set $V \subset \mathbb{A}^n$ is *reducible* if there exist two algebraic sets V_1 and V_2 in $\mathbb{A}^n$ and $V_i \neq V, i = 1, 2$ such that $V = V_1 \cup V_2$. Otherwise V is *irreducible* if nonempty.

Proposition 1.9. *An algebraic set V is irreducible if and only if $I(V)$ (which is a radical ideal) is a prime ideal.*

Usual arguments involving noetherian induction show that:

Proposition 1.10. *Let V be an algebraic set in $\mathbb{A}^n$. Then there are unique irreducible algebraic sets $V_1, \ldots, V_m$ such that $V = V_1 \cup \cdots \cup V_m$ and $V_i \not\subset V_j$ for all $i \neq j$. The V_i are called the irreducible components of V.*

This prompts the following definition.

Definition 1.11. Assume k is algebraically closed. Then an affine algebraic set V is called an affine *variety* if $I(V)$ is a prime ideal in $k[X_1, \ldots, X_n]$, i.e. V is an irreducible affine algebraic set.

Note that varieties are defined over algebraically closed fields. The issue is that algebraic sets over fields that are not algebraically closed can contain fewer "visible" points, so that the geometry does not contain as much information as the algebra. For instance, over the field of real numbers, $x^2 + y^2 = 0$ and $x^4 + y^4 = 0$ have the same visible points, but clearly these two should correspond to different objects. This is one reason

why scheme theory becomes specially important when dealing with objects over non-algebraically closed fields. (This is not to say that scheme theory doesn't become important over algebraically closed fields ...)

If we are given an algebraic set V over an algebraically closed field, the previous proposition and discussion gives a criterion for telling whether it is irreducible or not. We also need a way to describe V in terms of a given set of polynomials that define V. An important result, *Nullstellensatz*, or zeroes theorem, gives us the precise description between ideals and algebraic sets. We emphasize that this is also over algebraically closed fields.

Theorem 1.12 (Weak Nullstellensatz)**.** *Let k be an algebraically closed field. If I is a proper ideal in $k[X_1, \ldots, X_n]$, then $V(I) \neq \emptyset$.*

Theorem 1.13 (Strong Nullstellensatz)**.** *Let k be an algebraically closed field. Let I be an ideal in $k[X_1, \ldots, X_n]$. Then $I(V(I)) = \mathrm{Rad}(I)$.*

To see why this is a particularly important result, note the following corollary:

Corollary 1.14. *Let k be algebraically closed. If I is a prime ideal, then $V(I)$ is irreducible. There is a one-to-one correspondence between prime ideals and irreducible algebraic sets. Maximal ideals correspond to points.*

You might be tempted to think that we already proved in Proposition 1.9 that an irreducible algebraic set corresponds to prime ideals, but that is not correct! We showed that if we *start with* an algebraic set V, then $I(V)$ is prime if and only if V is irreducible. But to go the other way- to start with a prime ideal P and proving that it actually corresponds to a non-empty variety - requires Nullstellensatz. And this breaks down over non-algebraically closed fields. As an example, take the ideal $(x^2 + 1)$ in $\mathbb{R}[x]$ which is prime, but it corresponds to the empty set of points in geometry and so we cannot reconstruct the ideal from its set of points because $I(\emptyset) = \mathbb{R}[x]$, not $(x^2 + 1)$. Weak Nullstellensatz over algebraically closed fields ensures that this cannot happen- for any proper ideal I, the algebraic subset $V(I)$ has to be nonempty- and then strong Nullstellensatz steps in for prime ideals P to show that $I(V(P)) = P$, completing the correspondence. Scheme theory addresses this problem by redefining what a 'point' is in terms of prime ideals directly.

2. Theory of algebraic sets over non-algebraically closed fields

Since we will in fact be working over non-algebraically closed fields for a lot of this course, it is important to establish some connection between geometry over such fields and geometry over their algebraic closures. In our context, this is done by utilising Galois theory.

Let K be a perfect field and let $\bar{K}$ be a fixed algebraic closure of K. Let G_K denote the Galois group $\mathrm{Gal}(\bar{K}/K)$.

Note that G_K acts on $\mathbb{A}^n(\bar{K})$ by acting on co-ordinates:

$$\sigma.(a_1, \ldots, a_n) = (\sigma(a_1), \ldots, \sigma(a_n)).$$

Thus, $\mathbb{A}^n(K)$ is seen to be the set of fixed points of this action.

$$\mathbb{A}^n(K) = \{P \in \mathbb{A}^n(\bar{K}) : P = \sigma(P) \text{ for all } \sigma \in G_K\}.$$

Note that for any algebraic set V defined over K, we get that the *set of K-rational points of V* can be written as $V(K) = V(\bar{K}) \bigcap \mathbb{A}^n(K)$.

Definition 2.1. For V an algebraic set defined over $\bar{K}$, let us define

$$I(V/K) = \{f \in K[X_1, \ldots, X_n] : f(P) = 0 \text{ for all } P \in V\} = I(V) \bigcap K[X_1, \ldots, X_n].$$

Lemma 2.2. *If V is an algebraic set defined over $\bar{K}$, then V is defined over K, i.e. as the vanishing set of polynomials with coefficients in K, if and only if $I(V) = I(V/K)\bar{K}[X_1, \ldots, X_n]$.*

Remark 2.3. Just to emphasize the point once again, it is NOT enough to check whether $I(V/K)$ is a prime ideal to check whether V is a variety. You can take the $(x^2 + 1)$ example again to see this.

Notice that if $f(X_1, \ldots, X_n) \in K[X_1, \ldots, X_n]$ and $P \in \mathbb{A}^n(\bar{K})$, then for any $\sigma \in G_K$, we have $f(\sigma(P)) = \sigma(f(P))$. Hence, if V is defined over K, the G_K-action on $\mathbb{A}^n(\bar{K})$ restricts to $V(\bar{K})$, and clearly

$$V(K) = \{P \in V(\bar{K}) : \sigma(P) = P \text{ for all } \sigma \in G_K\}.$$

Given an affine variety V, we want to understand functions on it. The definition below provides us with a useful theory of such functions.

Definition 2.4 (Affine coordinate ring)**.** Let V be a variety defined over K, i.e. V is defined over K and V is an irreducible algebraic set over $\bar{K}$. Then the affine coordinate ring of V over K is defined to be:

$$K[V] = \frac{K[X_1, \ldots, X_n]}{I(V/K)}.$$

It is an exercise in ring theory to see that this is an integral domain. Its fraction field is denoted by $K(V)$ and is called the *function field of* V/K. Similarly, we define $\bar{K}[V]$ and $\bar{K}(V)$ by replacing K by $\bar{K}$.

Note that any element $f \in \bar{K}[V]$ gives a well-defined function $f : V \to \bar{K}$, since the ideal we quotient out by is precisely the ideal that vanishes on V. If $f \in \bar{K}[X_1, \ldots, X_n]$, G_K acts on it by acting on its coefficients. Hence if V is defined over K, G_K maps $I(V)$ to itself under this action, and so G_K acts on $\bar{K}[V]$ and $\bar{K}(V)$ respectively. It is an exercise to see that $K[V]$ and $K(V)$ are fixed points under this action. Note that for any f and P, we have $\sigma(f(P)) = (\sigma(f))(\sigma(P))$.

3. Dimension and smoothness

In this section, we describe further properties of varieties such as dimension and smoothness. For the same reasons as above (spectre of missing points haunting us...), these are best made over an algberaically closed field, even if the underlying variety is defined over a smaller subfield.

Definition 3.1. Let V be a variety. The *dimension* of V is the transcendence degree of $\bar{K}(V)$ over $\bar{K}$.

Note in particular that if $V \subset \mathbb{A}^n$ is given by a single nonconstant polynomial equation $f(X_1, \ldots, X_n) = 0$, then $\dim V = n - 1$. In fact, it is a good exercise to see that the converse is also true.

Definition 3.2. Let V be a variety, $P \in V$, and $f_1, \ldots, f_m$ be a set of generators for $I(V) \subset \bar{K}[X_1, \ldots, X_m]$. Then V is *smooth* or *nonsingular* at P if the $m \times n$ matrix

$$\left(\frac{\partial f_i}{\partial X_j}(P)\right)_{1 \le i \le m 1 \le j \le n}$$

has rank $= n - \dim V$. If V is smooth at every point, we say V is *smooth* or *nonsingular*.

This is the usual definition of smoothness coming from differential geometry and tangent spaces. There is another characterisation of smoothness which is quite useful. This characterisation is not that different than the previous one; as it is essentially about the linear dual of the tangent space at P.

Definition 3.3. For each point $P \in V$, we define an ideal M_P of $\bar{K}[V]$ by

$$M_P = \{f \in \bar{K}[V] : f(P) = 0\}.$$

This is a maximal ideal because it is the kernel of the evaluation-at-P homomorphism from $\bar{K}[V] \to \bar{K}$.

By Hilbert basis theorem, we get that M_P/M_P^2 is a finite dimensional vector space over $\bar{K}$.

Proposition 3.4. *Let V be a variety. A point $P \in V$ is smooth if and only if*

$$\dim_{\bar{K}} M_P/M_P^2 = \dim V.$$

We end with another definition. It tells us what are 'functions on V at P'.

Definition 3.5. Let V be a variety and point $P \in V$.*The local ring of V at P*, denoted $\bar{K}[V]_P$, is defined as the localization of $\bar{K}[V]$ at M_P.

If $F \in \bar{K}[V]_P$, then $F = f/g$ for some $f, g \in \bar{K}[V]$ with $g(P) \neq 0$. In particular, $F(P) = \frac{f(P)}{g(P)}$ is well-defined at P. These functions are called *regular at P*.